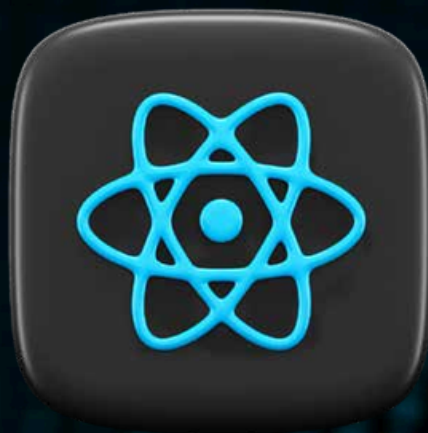


React Native Full Stack Development



Course Curriculum

Introduction to React Native Development

Overview of Mobile Development

- Introduction to mobile app development and the significance of cross-platform development.
- React Native: A framework to build mobile applications for iOS and Android using a single codebase.
- Differences between React Native and Native Development (iOS/Android).
- Overview of Mobile App Development: Native vs Hybrid vs Cross-Platform.
- Role of React Native and Firebase in Full-Stack Mobile App Development.
- Setting Up the Development Environment: Visual Studio Code, React Native CLI, Expo, and Firebase.

Front-End Development

Module-1 : HTML, CSS, andBootstrap

- HTML Fundamentals: Structure of an HTML Document, Semantic HTML5, Forms, Tables, and Multimedia
- CSS Basics: Selectors, Box Model, Flexbox, Grid Layout, Styling Forms, Animations, and Transitions
- Bootstrap: Grid System and Breakpoints, Components (Navbar, Cards, Buttons, Forms), Customizing Themes
- Responsive Design: Using Media Queries, Mobile-First Design Principles.

Module-2 : JavaScript (Pre-requisite)

- JavaScript Basics : Variables (var, let, const), Data Types (String, Number, Boolean, Undefined, Null), Type Conversion
- Control Flow : Conditionals (if, else, switch), Loops (for, while, do while)
- Functions and Objects : Declaring Functions, Arrow Functions, Objects, Arrays, and Array Methods
- Advanced JavaScript Concepts : ES6 Features, Destructuring, Template Literals, Spread/Rest Operators, Arrow Functions, this keyword
- Asynchronous JavaScript : Callbacks, Promises, async/await, Closures, Higher-Order Functions

Module-3: React Native Basics

- Introduction to React Native: React Native Overview, Setting up a React Native Project with Expo or React Native CLI.
- Components and JSX: Understanding Functional and Class Components, JSX Syntax.
- State and Props: Managing state in components, passing data through props.
- Handling User Input: Forms, Text Inputs, Buttons, Handling Events
- Styling in React Native: Using StyleSheet, Flexbox, and responsive design for mobile apps

Module-4: React Navigation & State Management

- React Navigation: Setting up Stack, Tab, and Drawer Navigation.
- React Navigation: Passing Params, Deep Linking, and Nested Navigators.
- State Management: Introduction to Context API and Redux for global state management.
- Handling Async Operations: Making HTTP requests with fetch or Axios for interacting with APIs.

Module-5: Firebase Backend Integration

- Introduction to Firebase: Firebase Authentication, Firestore Database, Firebase Storage.
- Firebase Authentication: User Sign Up, Login, Logout, Password Reset.
- Firebase Firestore: Cloud Firestore Basics, Reading, Writing, and Updating Data.
- Firebase Storage: Uploading and retrieving files (Images, Videos).
- Real-Time Database: Implementing Real-Time Data Synchronization with Firebase.

Module-6: Deployment of React Native Mobile App

- Deploying React Native Apps: Using Expo or React Native CLI for app builds and publishing to the App Store and Google Play.
- Firebase Hosting: Deploying static sites and front-end applications using Firebase Hosting for web applications if required.

Module-7: Capstone Project

- **Project Objective:** Build a Full-Stack Mobile Application using React Native (Front-End) and Firebase (Backend-less data storage).
- **Example Projects:** E-Commerce App, Social Media App, Task Management App, or Personal Finance Tracker.
- **Final Deliverables:** A working mobile app, Firebase integration, and deployment on Google Play/Apple App Store.

Course Duration

- **Total Duration:** 12 to 16 weeks
- **React Native Development:** 8-10 weeks
- **Firebase Integration and Real-Time Data:** 3-4 weeks
- **Project Development:** 2-4 weeks

**Scan to Start Your
IT Career at JZ**

